

Quiz 1 — Solution Sheet

1. Which method for which question

Read the question, find its row, use that method. Every question in this quiz is one of these nine types.

THE QUESTION ASKS	DO THIS	USED IN
Compute AB	Row of $A \times$ column of B	Q1
Compute Ax	Each row of $A \times$ the vector x	Q5
Compute $2A - B$	Entry by entry, position with position	Q7
Find k so vectors are dependent	Vectors as columns $\rightarrow \det = 0 \rightarrow$ solve for k	Q2
Which vector is in $\text{span}\{v_1, v_2\}$	Build $\alpha v_1 + \beta v_2 \rightarrow$ get a rule between coordinates $\rightarrow$ test options	Q6
Find a, b with $av_1 + bv_2 = \text{target}$	Split into 2 equations $\rightarrow$ eliminate (or Cramer)	Q9
Find $\dim(W)$	$\dim(W) = n - m$	Q3, Q4, Q10, Q11, Q12, Q14
Which set is a subspace	3 tests: origin, addition, scaling	Q8, Q16
Which matrix is PSD / PD	Symmetry first, then diagonal, then det	Q15, Q17, Q18
Find k for infinitely many solutions	$\det = 0$ and the two equations are the same line	Q13
Solve 2 equations in 2 unknowns	Add/subtract to kill one unknown, then back-substitute	Q19

2. Symbols

SYMBOL	MEANING
$\mathbb{R}^2, \mathbb{R}^3$	vectors made of 2 numbers / 3 numbers
x_1, x_2, x_3	coordinate 1, 2, 3 of one vector x
$u = (u_1, u_2), w = (w_1, w_2)$	two members of a set. Letter = which vector, subscript = which coordinate
v_1, v_2, v_3	whole vectors, named by the question itself
a_{ij}, b_{ij}	matrix entry in row i , column j
a, b, c, d	the four entries of a general 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

SYMBOL	MEANING
$c, s, t, k, \alpha, \beta$	plain single numbers
x^T	column written sideways: $(1, 0, 1)^T$ is a column
inside { }	"such that"
n	how many numbers each vector carries (size of the surrounding space)
m	how many independent rules (equations) are given
$\dim(W)$	how many coordinates you may still choose freely
$\det(A)$	one number from a square matrix. $\det = 0 \Rightarrow$ columns are dependent
span	every vector you can build by stretching and adding the given ones
subspace	contains the origin, and never escapes under $+$ or $\times$ number
PSD / PD	symmetric, plus ≥ 0 conditions / plus > 0 conditions

3. $\dim(W) = n - m$, and what m means

- n = how many numbers each vector has. $\mathbb{R}^2 \rightarrow n = 2$. $\mathbb{R}^3 \rightarrow n = 3$.
- m = how many **independent** rules. Count the "=" signs, then delete any equation that is another one multiplied by a number or made by adding others.
- $\dim(W) = n - m$ = how many coordinates are still free to choose.

m in four cases (all inside $\mathbb{R}^3$, so $n = 3$):

RULES GIVEN	M	WHY	DIM	SHAPE
$x_1 + x_2 + x_3 = 0$	1	one "=" sign	2	plane
$x_1 = 0, x_2 = 0$	2	two different rules	1	line
$x_1 + x_2 = 0, 2x_1 + 2x_2 = 0$	1	second = $2 \times$ first, so it is not new	2	plane
$x_1 = 0, x_2 = 0, x_3 = 0$	3	three different rules	0	origin only

Two rules that are not equations:

- $x \geq 0$ has no "=" sign, so it is not a constraint equation. Such sets are usually not subspaces.
- The right side must be 0. $x_1 + x_2 = 5$ is not a subspace, so $\dim = n - m$ does not apply.

Q1

Question. $A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 4 & 1 \end{pmatrix}$. Find $C = AB$.

Spot it. Two matrices multiplied together $\rightarrow$ row $\times$ column.

METHOD

1. Check sizes: columns of A must equal rows of B . Answer size = rows of $A \times$ columns of B .
2. Use $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j}$ — take **row i of A** and **column j of B** , multiply matching numbers, add.
3. Fill in all four entries.

WORKING

STEP	RESULT
Sizes	A is 2×2 , B is $2 \times 2 \rightarrow$ allowed, answer is 2×2
$c_{11} = \text{row 1} \times \text{col 1}$	$(1)(2) + (2)(4) = 2 + 8 = 10$
$c_{12} = \text{row 1} \times \text{col 2}$	$(1)(0) + (2)(1) = 0 + 2 = 2$
$c_{21} = \text{row 2} \times \text{col 1}$	$(-1)(2) + (3)(4) = -2 + 12 = 10$
$c_{22} = \text{row 2} \times \text{col 2}$	$(-1)(0) + (3)(1) = 0 + 3 = 3$

$$AB = \begin{pmatrix} 10 & 2 \\ 10 & 3 \end{pmatrix}$$

ANSWER

$$\text{(b)} \quad \begin{pmatrix} 10 & 2 \\ 10 & 3 \end{pmatrix}$$

Trap. $AB \neq BA$. Row always from the **first** matrix, column always from the **second**.

Q2

Question. Find k so that $v_1 = (1, 0, 1)^T$, $v_2 = (0, 1, 1)^T$, $v_3 = (1, k, 2)^T$ are linearly dependent.

Spot it. "Dependent" + as many vectors as coordinates $\rightarrow \det = 0$.

METHOD

1. Stand the vectors up as **columns** of a matrix M .
2. Expand $\det(M)$ along the row or column with the most zeros. Signs go $+$ $-$ $+$.
3. Set $\det(M) = 0$ and solve for k .

Small determinant needed in step 2: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.

WORKING

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & k \\ 1 & 1 & 2 \end{pmatrix}$$

STEP	RESULT
Expand along row 1	$1 \cdot \begin{vmatrix} 1 & k \\ 1 & 2 \end{vmatrix} - 0 \cdot \begin{vmatrix} 0 & k \\ 1 & 2 \end{vmatrix} + 1 \cdot \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}$
First small det	$(1)(2) - (k)(1) = 2 - k$
Middle term	$\times 0$, so it disappears
Third small det	$(0)(1) - (1)(1) = -1$
Add up	$\det(M) = (2 - k) - 0 + (-1) = 1 - k$
Set to zero	$1 - k = 0 \implies k = 1$

ANSWER

$$\text{(a) } k = 1$$

Trap. The signs alternate $+$ $-$ $+$ across the top row. Expanding along a row with a zero saves one whole calculation.

Q3

Question. $W = \{x \in \mathbb{R}^2 \mid x_1 = 2x_2\}$. Find $\dim(W)$.

Spot it. Asks for a dimension $\rightarrow \dim(W) = n - m$.

METHOD (USED AGAIN IN Q4, Q10, Q11, Q12, Q14)

- 1. **Count** n = how many numbers each vector has.
- 2. **Rewrite each rule with 0 on the right**, then **count** m = how many independent rules.
- 3. $\dim(W) = n - m$.
- 4. **Check:** call the free coordinates s, t , write x as a sum of fixed vectors. The number of those vectors must equal your answer.

WORKING

STEP	RESULT
1. n	vectors are $(x_1, x_2) \rightarrow n = 2$
2. rule and m	$x_1 - 2x_2 = 0 \rightarrow m = 1$
3. dimension	$\dim(W) = 2 - 1 = 1$
4. check	free: $x_2 = t$; forced: $x_1 = 2t$; so $x = t \begin{pmatrix} 2 \\ 1 \end{pmatrix} \rightarrow 1$ vector ✓

ANSWER

(b) 1

Picture. A line through the origin. One direction to walk in = dimension 1.

Q4

Question. $W = \{x \in \mathbb{R}^3 \mid x_1 = 2x_2 + x_3\}$. Find $\dim(W)$.

Spot it. Dimension $\rightarrow$ **same 4 steps as Q3.**

WORKING

STEP	RESULT
1. n	vectors are $(x_1, x_2, x_3) \rightarrow n = 3$
2. rule and m	$x_1 - 2x_2 - x_3 = 0 \rightarrow m = 1$
3. dimension	$\dim(W) = 3 - 1 = 2$
4. check	free: $x_2 = s, x_3 = t$; forced: $x_1 = 2s + t$

$$x = \begin{pmatrix} 2s + t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \rightarrow \mathbf{2 \text{ vectors } \checkmark}$$

The two vectors are not multiples of each other, so they are independent and form a basis.

ANSWER

(d) 2

Picture. A plane through the origin. Two directions to slide in = dimension 2.

Q5

Question. $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$, $x = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. Find Ax .

Spot it. Matrix times a vector $\rightarrow$ each row meets the whole vector.

METHOD

1. Row 1 of A against $x \rightarrow$ first number of the answer.
2. Row 2 of A against $x \rightarrow$ second number of the answer.
3. Stack them into a column.

WORKING

STEP	RESULT
Row 1	$y_1 = (2)(2) + (-1)(-1) = 4 + 1 = 5$
Row 2	$y_2 = (3)(2) + (4)(-1) = 6 - 4 = 2$

$$Ax = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

ANSWER

$$\text{(c)} \quad \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

Trap. $(-1)(-1) = +1$. Minus times minus is plus.

Q6

Question. Which vector is in $\text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$? Options: **(a)** $(1, 1, 1)^T$, **(b)** $(2, -1, 1)^T$, **(c)** $(2, -1, 3)^T$, **(d)** $(0, 2, 1)^T$.

Spot it. "In the span" $\rightarrow$ turn the span into one rule between the coordinates, then test the options.

METHOD

1. Build the general member: $\alpha v_1 + \beta v_2$.
2. Read off each coordinate. Replace α and β by the coordinates they equal.
3. What is left is one rule. Test every option against it.

WORKING

$$\alpha \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \\ \alpha + \beta \end{pmatrix} \rightarrow x_1 = \alpha, x_2 = \beta, x_3 = \alpha + \beta$$

$$\text{Rule: } x_3 = x_1 + x_2$$

OPTION	$x_1 + x_2$	x_3	EQUAL?
(a) $(1, 1, 1)$	2	1	no
(b) $(2, -1, 1)$	1	1	yes
(c) $(2, -1, 3)$	1	3	no
(d) $(0, 2, 1)$	2	1	no

Confirm (b) by building it: $2v_1 + (-1)v_2 = (2, -1, 1)^T \checkmark$

ANSWER

(b) $(2, -1, 1)^T$

Q7

Question. $A = \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}$. Find $2A - B$.

Spot it. Only $+$, $-$ and a number in front $\rightarrow$ work entry by entry. No row $\times$ column here.

METHOD

1. Multiply every entry of A by 2.
2. Subtract B position by position: row 1 col 1 with row 1 col 1, and so on.

WORKING

STEP	RESULT
$2A$	$\begin{pmatrix} 2 & -4 \\ 6 & 0 \end{pmatrix}$
top-left	$2 - 4 = -2$
top-right	$-4 - 1 = -5$
bottom-left	$6 - (-1) = 6 + 1 = 7$
bottom-right	$0 - 2 = -2$

$$2A - B = \begin{pmatrix} -2 & -5 \\ 7 & -2 \end{pmatrix}$$

ANSWER

$$\text{(d)} \quad \begin{pmatrix} -2 & -5 \\ 7 & -2 \end{pmatrix}$$

Trap. Bottom-left is $6 - (-1) = 7$, not 5.

Q8

Question. Which is a subspace of $\mathbb{R}^2$? **(a)** $x + y = 1$, **(b)** $xy = 0$, **(c)** $x - 2y = 0$, **(d)** $x \geq 0$.

Spot it. "Which set is a subspace" $\rightarrow$ run the 3 tests.

METHOD

Write a member as $u = (u_1, u_2)$ and a second as $w = (w_1, w_2)$. Let c be any number.

1. Origin: does $(0, 0)$ obey the rule?

2. Addition: is $u + w = (u_1 + w_1, u_2 + w_2)$ still in the set?

3. Scaling: is $cu = (cu_1, cu_2)$ still in the set, **for every** c , including negatives?

Stop at the first test that fails. **To fail** you only need one example with real numbers. **To pass** you must use letters, because it has to hold for every member.

Fast filter: cross out any rule with a non-zero right side, any $\geq$ or $\leq$, and any two variables multiplied together.

WORKING

OPTION	1. ORIGIN	2. ADDITION	3. SCALING	VERDICT
(a) $x + y = 1$	$0 + 0 = 0 \neq 1$ fails	—	—	not a subspace
(b) $xy = 0$	$(0)(0) = 0 \checkmark$	$u = (1, 0), w = (0, 1) \rightarrow u + w = (1, 1), (1)(1) = 1 \neq 0$ fails	—	not a subspace
(c) $x - 2y = 0$	$0 - 2(0) = 0$ $\checkmark$	$\checkmark$ (letters below)	$\checkmark$ (letters below)	subspace
(d) $x \geq 0$	$0 \geq 0 \checkmark$	$u_1 + w_1 \geq 0 \checkmark$	$u = (1, 0), c = -1 \rightarrow (-1, 0), -1 \geq 0$ false fails	not a subspace

Proof for (c) with letters. Members obey $u_1 - 2u_2 = 0$ and $w_1 - 2w_2 = 0$.

$$\text{Addition: } (u_1 + w_1) - 2(u_2 + w_2) = (u_1 - 2u_2) + (w_1 - 2w_2) = 0 + 0 = 0 \checkmark$$

$$\text{Scaling: } cu_1 - 2cu_2 = c(u_1 - 2u_2) = c \cdot 0 = 0 \checkmark$$

ANSWER

$$\text{(c) } \{(\mathbf{x}, \mathbf{y}) : \mathbf{x} - 2\mathbf{y} = \mathbf{0}\}$$

Q9

Question. Find (a, b) with $a \begin{pmatrix} 1 \\ 2 \end{pmatrix} + b \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$.

Spot it. Find the two mixing amounts $\rightarrow$ split into 2 equations and solve.

METHOD

1. Match the top numbers of both sides, then the bottom numbers. That gives 2 equations.
2. Solve by elimination, or by Cramer's rule below.
3. Put the answer back in to check.

Cramer for $\begin{cases} c_{11}a + c_{12}b = d_1 \\ c_{21}a + c_{22}b = d_2 \end{cases}$:

$$D = c_{11}c_{22} - c_{12}c_{21}, \quad a = \frac{d_1c_{22} - d_2c_{12}}{D}, \quad b = \frac{c_{11}d_2 - c_{21}d_1}{D}$$

(The coefficients are called c_{ij} and the right side d_i only because the letters a and b are already used by the two unknowns. In Q13 and Q19, where the unknowns are x and y , they are called a_{ij} and b_i as usual.)

WORKING

$$\begin{cases} a + 2b = 5 \\ 2a - b = 0 \end{cases} \quad c_{11} = 1, c_{12} = 2, d_1 = 5, \quad c_{21} = 2, c_{22} = -1, d_2 = 0$$

STEP	RESULT
D	$(1)(-1) - (2)(2) = -1 - 4 = -5$
a	$\frac{(5)(-1) - (0)(2)}{-5} = \frac{-5}{-5} = 1$
b	$\frac{(1)(0) - (2)(5)}{-5} = \frac{-10}{-5} = 2$
check	$1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix} \checkmark$

ANSWER

(c) $(a, b) = (1, 2)$

Faster in the exam. Put each option into the equation. $(1, 2)$ works in about 10 seconds.

Q10

Question. $W = \{x \in \mathbb{R}^3 \mid x_1 - x_2 = x_3\}$. Find $\dim(W)$.

Spot it. Dimension $\rightarrow$ **same 4 steps as Q3.**

WORKING

STEP	RESULT
1. n	$n = 3$
2. rule and m	move all to one side: $x_1 - x_2 - x_3 = 0 \rightarrow m = 1$
3. dimension	$\dim(W) = 3 - 1 = 2$
4. check	free: $x_2 = s, x_3 = t$; forced: $x_1 = s + t$

$$x = \begin{pmatrix} s+t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \rightarrow \mathbf{2 \text{ vectors } \checkmark}$$

ANSWER

(d) 2

Step 2 matters. Always push everything to one side first, so the rule reads " $\dots = 0$ ", then count.

Q11

Question. $W = \{x \in \mathbb{R}^3 \mid x_1 = 0\}$. Find $\dim(W)$.

Spot it. Dimension $\rightarrow$ **same 4 steps as Q3.**

WORKING

STEP	RESULT
1. n	$n = 3$
2. rule and m	$x_1 = 0 \rightarrow m = 1$
3. dimension	$\dim(W) = 3 - 1 = 2$
4. check	free: $x_2 = s, x_3 = t$; forced: $x_1 = 0$

$$x = s \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \mathbf{2 \text{ vectors } \checkmark}$$

ANSWER

(d) 2

Note. A rule like $x_1 = 0$ costs exactly **one** freedom, the same as any other single equation.

Q12

Question. $W = \{x \in \mathbb{R}^2 \mid x_1 = 0, x_2 \in \mathbb{R}\}$. Find $\dim(W)$.

Spot it. Dimension $\rightarrow$ **same 4 steps as Q3.**

WORKING

STEP	RESULT
1. n	$n = 2$
2. rule and m	$x_1 = 0 \rightarrow m = 1$
3. dimension	$\dim(W) = 2 - 1 = 1$
4. check	free: $x_2 = t \rightarrow x = t \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow$ 1 vector ✓

ANSWER

(b) 1

Compare with Q11. Same rule $x_1 = 0$, but $\dim = 2$ there and $\dim = 1$ here, because n changed. **Always check which space you are in first.**

Q13

Question. Find k so the system has **infinitely many** solutions: $\begin{cases} kx + y = 2 \\ 2x + 2y = 4 \end{cases}$

Spot it. "Infinitely many solutions" $\rightarrow$ two-part test, not one.

METHOD

1. $\det(A) = 0$, where A holds the coefficients. This makes the two lines **parallel or identical**.
2. Then check the whole equations match (divide one by a number and compare). This separates *identical* (infinitely many) from *parallel* (no solution).

WORKING

$$A = \begin{pmatrix} k & 1 \\ 2 & 2 \end{pmatrix}$$

STEP	RESULT
1. $\det(A)$	$(k)(2) - (1)(2) = 2k - 2$
set to 0	$2k - 2 = 0 \implies k = 1$
2. match check	equation 2 $\div$ 2 gives $x + y = 2$; with $k = 1$ equation 1 is $x + y = 2$ — identical ✓

ANSWER

(d) $k = 1$

Trap. $\det = 0$ on its own could equally mean **no** solution. Only the matching right-hand sides prove "infinitely many".

Q14

Question. $W = \{x \in \mathbb{R}^2 \mid x_2 = 2x_1\}$. Find $\dim(W)$.

Spot it. Dimension $\rightarrow$ **same 4 steps as Q3.**

WORKING

STEP	RESULT
1. n	$n = 2$
2. rule and m	$x_2 - 2x_1 = 0 \rightarrow m = 1$
3. dimension	$\dim(W) = 2 - 1 = 1$
4. check	free: $x_1 = t$; forced: $x_2 = 2t \rightarrow x = t \begin{pmatrix} 1 \\ 2 \end{pmatrix} \rightarrow$ 1 vector ✓

ANSWER

(d) 1

Note. Q3 was $x_1 = 2x_2$ and this is $x_2 = 2x_1$. One equation in $\mathbb{R}^2$ always leaves 1 freedom. Only the direction of the line changes.

Q15

Question. Which matrix is **Positive Semi-Definite (PSD)**? (a) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, (b) $\begin{pmatrix} 1 & 3 \\ 2 & 8 \end{pmatrix}$, (c) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, (d) $\begin{pmatrix} 1 & 2 \\ 3 & 8 \end{pmatrix}$.

Spot it. PSD $\rightarrow$ three checks, in this order. Symmetry first, because it kills options in two seconds.

METHOD

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

1. Symmetric? $b = c$. If not, stop — it cannot be PSD.

2. Diagonal? $a \geq 0$ and $d \geq 0$.

3. Determinant? $ad - bc \geq 0$.

Everything uses $\geq$, so **zeros are allowed**.

WORKING

OPTION	1. $B = C$?	2. $A \geq 0, D \geq 0$?	3. $\text{DET} \geq 0$?	VERDICT
(a) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	$0 = 0$ ✓	$a = -1 < 0$ fails	—	not PSD
(b) $\begin{pmatrix} 1 & 3 \\ 2 & 8 \end{pmatrix}$	$3 \neq 2$ fails	—	—	not PSD
(c) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$0 = 0$ ✓	$0 \geq 0, 0 \geq 0$ ✓	$0 \geq 0$ ✓	PSD
(d) $\begin{pmatrix} 1 & 2 \\ 3 & 8 \end{pmatrix}$	$2 \neq 3$ fails	—	—	not PSD

ANSWER

$$(c) \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Why the zero matrix counts. It gives $x^T A x = 0$ always, and zero is not negative. This is exactly the case PSD allows and PD (Q17) forbids.

Q16

Question. Which is a subspace of $\mathbb{R}^2$? **(a)** $y = 1$, **(b)** $x + y = 0$, **(c)** $x \geq 0$, **(d)** $x - y = 1$.

Spot it. Subspace $\rightarrow$ **same 3 tests as Q8**, same letters $u = (u_1, u_2)$, $w = (w_1, w_2)$, same order.

WORKING

OPTION	1. ORIGIN	2. ADDITION	3. SCALING	VERDICT
(a) $y = 1$	$0 \neq 1$ fails	—	—	not a subspace
(b) $x + y = 0$	$0 + 0 = 0$ ✓	✓ (letters below)	✓ (letters below)	subspace
(c) $x \geq 0$	$0 \geq 0$ ✓	$u_1 + w_1 \geq 0$ ✓	$u = (1, 0)$, $c = -1 \rightarrow (-1, 0)$, $-1 \geq 0$ false fails	not a subspace
(d) $x - y = 1$	$0 - 0 = 0 \neq 1$ fails	—	—	not a subspace

Proof for (b) with letters. Members obey $u_1 + u_2 = 0$ and $w_1 + w_2 = 0$.

Addition: $(u_1 + w_1) + (u_2 + w_2) = (u_1 + u_2) + (w_1 + w_2) = 0 + 0 = 0$ ✓

Scaling: $cu_1 + cu_2 = c(u_1 + u_2) = c \cdot 0 = 0$ ✓

ANSWER

(b) $\{(\mathbf{x}, \mathbf{y}) : \mathbf{x} + \mathbf{y} = \mathbf{0}\}$

One-line filter. A line $ax + by = c$ is a subspace **only when** $c = 0$.

Q17

Question. Which matrix is **Positive Definite (PD)**? **(a)** $\begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$, **(b)** $\begin{pmatrix} 1 & 3 \\ 2 & 8 \end{pmatrix}$, **(c)** $\begin{pmatrix} 7 & 0 \\ 0 & 25 \end{pmatrix}$, **(d)** $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

Spot it. PD $\rightarrow$ same three checks as Q15, but every " $\geq$ " becomes " $>$ ". **No zeros allowed.**

METHOD

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

1. **Symmetric?** $b = c$.
2. **Top-left corner?** $a > 0$.
3. **Determinant?** $ad - bc > 0$.

WORKING

OPTION	1. $B = C$?	2. $A > 0$?	3. $\text{DET} > 0$?	VERDICT
(a) $\begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$	✓	$-1 > 0$ false fails	$0 > 0$ false	not PD
(b) $\begin{pmatrix} 1 & 3 \\ 2 & 8 \end{pmatrix}$	$3 \neq 2$ fails	—	—	not PD
(c) $\begin{pmatrix} 7 & 0 \\ 0 & 25 \end{pmatrix}$	$0 = 0$ ✓	$7 > 0$ ✓	$175 > 0$ ✓	PD
(d) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	✓	$0 > 0$ false fails	$0 > 0$ false	not PD

ANSWER

$$\text{(c)} \begin{pmatrix} 7 & 0 \\ 0 & 25 \end{pmatrix}$$

PSD vs PD in one line. Same three checks. PSD uses $\geq$ (zeros fine), PD uses $>$ (no zeros). The zero matrix is PSD but not PD.

Q18

Question. Which matrix is **PSD**? **(a)** $\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$, **(b)** $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, **(c)** $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, **(d)** $\begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}$.

Spot it. PSD $\rightarrow$ **same 3 checks as Q15**. Three options are diagonal, so step 2 alone decides them.

Shortcut. For a diagonal matrix $\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$ the eigenvalues **are** a and d . PSD just means $a \geq 0$ and $d \geq 0$.

WORKING

OPTION	1. $B = C$?	2. $A \geq 0, D \geq 0$?	3. $\text{DET} \geq 0$?	VERDICT
(a) $\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$	$3 \neq 2$ fails	—	—	not PSD
(b) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$	✓	$1 \geq 0, 0 \geq 0$ ✓	$0 \geq 0$ ✓	PSD
(c) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	✓	$-1 < 0$ fails	—	not PSD (negative definite)
(d) $\begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}$	✓	$-2 < 0$ fails	—	not PSD (indefinite)

ANSWER

(b) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

Note. PSD but **not** PD, because $\text{det} = 0$ (one eigenvalue is exactly zero).

Q19

Question. Solve $\begin{cases} 2x + y = 7 \\ x - y = 2 \end{cases}$

Spot it. Two equations, two unknowns, and the y terms are $+y$ and $-y \rightarrow$ add them to kill y .

METHOD

1. Add or subtract the two equations so one unknown disappears.
2. Solve the single equation that remains.
3. Put that value back into either original equation to get the other unknown.

WORKING

STEP	RESULT
1. add them	$(2x + y) + (x - y) = 7 + 2 \implies 3x = 9$
2. solve	$x = 3$
3. back-substitute into $x - y = 2$	$3 - y = 2 \implies y = 1$
check in equation 1	$2(3) + 1 = 7 \checkmark$

Cross-check with the inverse ($\det = (2)(-1) - (1)(1) = -3$):

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{-3} \begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 7 \\ 2 \end{pmatrix} = -\frac{1}{3} \begin{pmatrix} -9 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \checkmark$$

ANSWER

$$\text{(b) } (x, y) = (3, 1)$$

Faster in the exam. Put each option into both equations. $(3, 1)$: $2(3) + 1 = 7 \checkmark$ and $3 - 1 = 2 \checkmark$.

Revision Sheet

ALL ANSWERS

Q	TYPE	ANSWER
1	matrix $\times$ matrix	(b) $\begin{pmatrix} 10 & 2 \\ 10 & 3 \end{pmatrix}$
2	dependent $\rightarrow \det = 0$	(a) $k = 1$
3	$\dim, \mathbb{R}^2$	(b) 1
4	$\dim, \mathbb{R}^3$	(d) 2
5	matrix $\times$ vector	(c) $\begin{pmatrix} 5 \\ 2 \end{pmatrix}$
6	span	(b) $(2, -1, 1)^T$
7	$2A - B$	(d) $\begin{pmatrix} -2 & -5 \\ 7 & -2 \end{pmatrix}$
8	subspace	(c) $x - 2y = 0$
9	mixing amounts	(c) $(1, 2)$
10	$\dim, \mathbb{R}^3$	(d) 2
11	$\dim, \mathbb{R}^3$	(d) 2
12	$\dim, \mathbb{R}^2$	(b) 1
13	infinitely many solutions	(d) $k = 1$
14	$\dim, \mathbb{R}^2$	(d) 1
15	PSD	(c) zero matrix
16	subspace	(b) $x + y = 0$
17	PD	(c) $\begin{pmatrix} 7 & 0 \\ 0 & 25 \end{pmatrix}$
18	PSD	(b) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$
19	solve 2×2 system	(b) $(3, 1)$

THE FIVE FLOWS

#	FLOW	STEPS
1	Dimension	count $n \rightarrow$ write rules as " $\dots = 0$ " and count $m \rightarrow \dim = n - m \rightarrow$ check with free variables

#	FLOW	STEPS
2	Subspace	origin $\rightarrow$ addition $\rightarrow$ scaling. Fail with numbers, pass with letters
3	Dependent vectors	vectors as columns $\rightarrow$ $\det = 0 \rightarrow$ solve
4	PSD / PD	$b = c \rightarrow$ diagonal $\rightarrow$ $\det$. PSD uses $\geq$, PD uses $>$
5	Two lines	$\det \neq 0$ one solution; $\det = 0$ and equations match $\rightarrow$ infinitely many; $\det = 0$ and they clash $\rightarrow$ none

FASTEST EXAM FILTERS

- **Subspace:** cross out non-zero right sides, any $\geq$ or $\leq$, and any xy .
- **PSD / PD:** check $b = c$ first — it usually removes two options instantly.
- Any "solve" or "find the pair" **question:** substitute the options rather than solving.